

When the points are coincident, P and Q will share the same coordinates, and they will usually be designated by the same letter with a subscript identifying the link to which they are attached. For example, if 3 and 4 are the links to which the coincident points are attached, if body B corresponds to link 5, and if the frame is 1, the velocity and acceleration equations can be written as

$$\mathbf{v}_{P_3} = \mathbf{v}_{P_4} + {}^5\mathbf{v}_{P_3/P_4} \quad (3.27)$$

and

$$\mathbf{a}_{P_3} = \mathbf{a}_{P_4} + {}^5\mathbf{a}_{P_3/P_4} + 2\boldsymbol{\omega}_5 \times {}^5\mathbf{v}_{P_3/P_4} \quad (3.28)$$

Once again, ${}^5\mathbf{a}_{P_3/P_4}$ is called the acceleration of P_3 relative to P_4 when the observer is in system 5. The term $2\boldsymbol{\omega}_5 \times {}^5\mathbf{v}_{P_3/P_4}$ is the Coriolis component of acceleration, and it can be written as $\mathbf{a}_{P_3/P_4}^C$. Eq. (3.28) can then be written as

$$\mathbf{a}_{P_3} = \mathbf{a}_{P_4} + {}^5\mathbf{a}_{P_3/P_4} + \mathbf{a}_{P_3/P_4}^C$$

for graphical analyses.

In planar motion, the Coriolis component is normal to ${}^5\mathbf{v}_{P_3/P_4}$ and has the magnitude $2(\omega_5)|{}^5\mathbf{v}_{P_3/P_4}|$. Its sense is obtained by imagining ${}^5\mathbf{v}_{P_3/P_4}$ to be rotating about its tail in the $\boldsymbol{\omega}_5$ direction. The direction of movement of the head of ${}^5\mathbf{v}_{P_3/P_4}$ gives the sense. To illustrate the manner in which Eqs. (3.27) and (3.28) are used in graphical linkage analysis, consider the following example.

EXAMPLE 3.1

Velocity and Acceleration Analysis of a Quick-Return Mechanism

Find the sliding velocities of the slide, the angular accelerations of links 3 and 4, and the acceleration of slide 5 for the quick-return mechanism of Fig. 3.10. The dimensions are as shown. Link 2 is driven with a constant angular velocity of 10 rpm CCW.

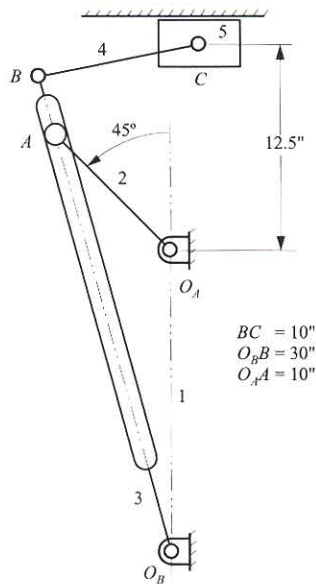


FIGURE 3.10 The quick return linkage to be analyzed in Example 3.1.

Solution

Link 2 is the driver, so we will begin the analysis with point A_2 . We must conduct the velocity analysis first. If that analysis is done carefully, we can proceed with the same points for the acceleration analysis. As in the previous examples, we will develop the basic equations first and then give the graphical procedure for solving them. The velocity of point A_2 is given by

$$\mathbf{v}_{A_2} = \mathbf{v}_{A_2/O_A} = \boldsymbol{\omega}_2 \times \mathbf{r}_{A_2/O_A}$$

In the analysis of mechanisms of this type, it is important to identify the link in which the observer is located. Therefore, the left superscripts will be maintained when the coordinate system is different from 1. We must now use the coincident point A_3 to be able to develop an equation relating a point on link 2 to a point on link 3. We can write the relative velocity equation in one of two ways:

$$\mathbf{v}_{A_2} = \mathbf{v}_{A_3} + \mathbf{v}_{A_2/A_3} = \mathbf{v}_{A_3} + {}^3\mathbf{v}_{A_2/A_3} \quad (3.29)$$

or

$$\mathbf{v}_{A_3} = \mathbf{v}_{A_2} + \mathbf{v}_{A_3/A_2} = \mathbf{v}_{A_2} + {}^2\mathbf{v}_{A_3/A_2} \quad (3.30)$$

To solve the problem, we must be able to recognize the direction of the relative velocity defined in the moving coordinate system. Referring to the mechanism in Fig. 3.10, we see that if the observer is fixed to link 2, it is not possible to identify directly the direction of the velocity ${}^2\mathbf{v}_{A_3/A_2}$; however, if the observer is in link 3, it is possible to identify the direction of the velocity ${}^3\mathbf{v}_{A_2/A_3}$ because the pin at A is constrained to move along the straight slot in link 3. Therefore, the direction of the velocity ${}^3\mathbf{v}_{A_2/A_3}$ must be along the slot. Because we can determine the direction of ${}^3\mathbf{v}_{A_2/A_3}$ by inspection, Eq. (3.29) is more useful than Eq. (3.30).

In problems such as this, it is important to identify clearly the links relative to which the velocity is usually measured. The same links can be used for the subsequent acceleration analysis, and it is usually

After Eq. (3.29) is solved for the unknowns, $\mathbf{v}_{A_3}$ will be known. Then $\mathbf{v}_{B_3}$ can be found from the velocity image of link 3 using O_B , A_3 , and B_3 . Knowing $\mathbf{v}_{B_3}$, which is the same as $\mathbf{v}_{B_4}$ we can write the following equation for the velocity of C_4 :

$$\mathbf{v}_{C_4} = \mathbf{v}_{B_4} + \mathbf{v}_{C_4/B_4} \quad (3.31)$$

Because the directions of $\mathbf{v}_{C_4}$ and $\mathbf{v}_{C_4/B_4}$ are known, we can solve Eq. (3.31) for the unknowns. After Eqs. (3.29) and (3.31) are solved, we can compute the angular velocities of links 3 and 4 from

$$\mathbf{v}_{B_3/O_B} = \boldsymbol{\omega}_3 \times \mathbf{r}_{B_3/O_B}$$

and

$$\mathbf{v}_{C_4/B_4} = \boldsymbol{\omega}_4 \times \mathbf{r}_{C_4/B_4}$$

For the acceleration analysis, we need only differentiate Eqs. (3.29) and (3.31). The results are

$$\mathbf{a}_{A_2} = \mathbf{a}_{A_3} + \mathbf{a}_{A_2/A_3}$$

and

$$\mathbf{a}_{C_4} = \mathbf{a}_{B_4} + \mathbf{a}_{C_4/B_4}$$

Expanding the equations in terms of vectors relative to moving coordinate systems, we obtain

$$\mathbf{a}_{A_2/O_A}^r + \mathbf{a}_{A_2/O_A}^t = \mathbf{a}_{A_3/O_B}^r + \mathbf{a}_{A_3/O_B}^t + {}^3\mathbf{a}_{A_2/A_3} + 2(\boldsymbol{\omega}_3 \times {}^3\mathbf{v}_{A_2/A_3}) \quad (3.32)$$

and

$$\mathbf{a}_{C_4} = \mathbf{a}_{B_3/O_B} + \mathbf{a}_{C_4/B_4}^r + \mathbf{a}_{C_4/B_4}^t \quad (3.33)$$

where

$$\mathbf{a}_{A_2/O_A}^r = r_{A_2/O_A} \omega_2^2 \text{ from } A \text{ to } O_A$$

$$\mathbf{a}_{A_2/O_A}^t = \alpha_2 \times r_{A_2/O_A} \text{ perpendicular to } \overrightarrow{AO_A}$$

$$\mathbf{a}_{A_3/O_B}^r = r_{A_3/O_B} \omega_2^2 \text{ from } A \text{ to } O_B$$

$$\mathbf{a}_{A_3/O_B}^t = \alpha_3 \times r_{A_3/O_B} (\alpha_3 \text{ is unknown but the result is perpendicular to } \overrightarrow{AO_B})$$

$${}^3\mathbf{a}_{A_2/A_3} \text{ has a magnitude that is unknown but its direction is along the slot in link 3}$$

$$2(\omega_3 \times {}^3\mathbf{v}_{A_2/A_3}) \text{ is the Coriolis acceleration perpendicular to } {}^3\mathbf{v}_{A_2/A_3}$$

$$\mathbf{a}_{C_4} \text{ is along the slider path of link 5}$$

$$\mathbf{a}_{B_3/O_B} \text{ is found by acceleration image of link 3}$$

$$\mathbf{a}_{C_4/B_4}^r = r_{C_4/B_4} \omega_4^2 \text{ from } C \text{ to } B$$

$$\mathbf{a}_{C_4/B_4}^t = \alpha_4 \times r_{C_4/B_4} \text{ has a magnitude that is unknown but a direction that is perpendicular to } \overrightarrow{CB}$$

Based on the position and velocity analyses, there will be only two unknown magnitudes in Eqs. (3.31) and (3.33). All of the directions will be known. Therefore, the equations can be solved.

Steps

1. Draw linkage to scale as shown in Fig. 3.11.
2. Compute v_{A_2} , and plot v_{A_2} normal to $\overrightarrow{OA}$ as $\overrightarrow{oa_2}$.
 $\omega_2 = 10 \times 2\pi/60 = 1.0472 \text{ rad/s CCW}$
 $v_{A_2} = \omega_2 \times r_{A_2/O_A} \times 10 = 10.472 \text{ in/s perpendicular to } \overrightarrow{OA} \text{ in the direction shown in Fig. 3.11.}$
3. Draw a line through a_2 parallel to $\overrightarrow{OB}$.
4. Draw a line through o normal to $\overrightarrow{BA_3}$. The intersection with the line from step 3 gives point a_3 .
5. Locate b_3 using the velocity image $\frac{ob_3}{oa_3} = \frac{OB}{OA}$.
 $ob_3 = 7.26 \times (30/26.48) = 8.22 \text{ in/s}$
6. Draw a line through b_3 normal to $\overrightarrow{BC}$.
7. Draw a line through o parallel to the slide. Its intersection with the line drawn in step 6 gives point c_4 .
8. Measure $v_{C_4} = oc_4$, $v_{A_2/A_3} = \overrightarrow{a_2a_3}$ and $v_{C_4/B_4} = \overrightarrow{c_4b_4}$.
 $v_{C_4} = 11.06 \text{ in/s}$, $v_{A_2/A_3} = 5.09 \text{ in/s}$, $v_{C_4/B_4} = 2.823 \text{ in/s}$
9. Compute $\omega_3 = v_{A_3/O_B}/r_{A_3/O_B}$ and $\omega_4 = v_{C_4/B_4}/r_{C_4/B_4}$.
 $\omega_3 = (9.07)/(26.37) = 0.344 \text{ rad/s CCW}$
 $\omega_4 = 2.823/5 = 0.565 \text{ rad/s CCW}$

14. Compute $\mathbf{a}_{A_3/O_B}^r$ and plot it from o' in the $\overrightarrow{A_3O_B}$ direction.

$$\mathbf{a}_{A_3/O_B}^r = \left| \mathbf{r}_{A_3/O_B} \right| \left| \boldsymbol{\omega}_3 \right|^2 = 26.37 \times 0.344^2 = 3.12 \text{ in/s}^2$$

15. Draw a line through the tip of $\mathbf{a}_{A_3/O_B}^r$ normal to $\overrightarrow{O_B A_3}$. This vector corresponds to $\mathbf{a}_{A_3/O_B}^t$. Its intersection with the line drawn in step 13 gives point a'_3 .

16. Locate point b'_3 using the acceleration image $\frac{o'b'_3}{o'a'_3} = \frac{O_B B}{O_B A}$.

$$o'b'_3 = 3.62 \times (30/26.37) = 4.12 \text{ in/s}^2$$

17. Plot $\mathbf{a}_{C_4/B_4}^r$ from point b'_3 parallel to $\overrightarrow{CB}$.

$$\mathbf{a}_{C_4/B_4}^r = \left| \mathbf{r}_{C_4/B_4} \right| \left| \boldsymbol{\omega}_4 \right|^2 = 5 \times 0.608^2 = 1.85 \text{ in/s}^2$$

18. Draw a line through the tip of $\mathbf{a}_{C_4/B_4}^r$ normal to $\overrightarrow{CB}$.

This vector corresponds to $\mathbf{a}_{C_4/B_4}^t$.

19. Draw a line through o' parallel to the slide. Its intersection with the line generated in step 18 gives point c'_4 .

20. Measure $\mathbf{a}_{A_3/O_B}^t$ and $\mathbf{a}_{C_4/B_4}^t$.

$$\mathbf{a}_{A_3/O_B}^t = 1.870 \text{ in/s}^2 \text{ and } \mathbf{a}_{C_4/B_4}^t = 3.197 \text{ in/s}^2$$

21. Compute $\alpha_3 = \frac{\mathbf{a}_{A_3/O_B}^t}{r_{A_3/O_B}}$ and $\alpha_4 = \frac{\mathbf{a}_{C_4/B_4}^t}{r_{C_4/B_4}}$ and get the senses of these angular accelerations by considering the directions of rotation needed to rotate the position vectors in the directions of $\mathbf{a}_{A_3/O_B}^t$ and $\mathbf{a}_{C_4/B_4}^t$, respectively.

$$\alpha_3 = 1.870 / (26.37) = 0.071 \text{ rad/s}^2 \text{ CW}$$

$$\alpha_4 = 3.197 / 5 = 0.640 \text{ rad/s}^2 \text{ CCW}$$

22. Measure $\mathbf{a}_{C_4}$.

$$\mathbf{a}_{C_4} = 0.7986 \text{ in/s}^2 \text{ directed to the right}$$

One of the useful features of the quick-return linkage is a long range of motion with relatively uniform velocity on the forward stroke. The small value of $\mathbf{a}_{C_4}$ is indicative of this property.

3.6 ROLLING CONTACT

Rolling contact is quite often used in practical linkages. In addition to the obvious case of a wheel rolling on a surface or a rail, rolling contact between a cam and a roller follower is a common example. Also, the pitch cylinders of spur and helical gear pairs and the pitch cones of bevel gear pairs can be considered to be in pure rolling contact. In that case, although the actual physical contact between the gear teeth is a general combination of rolling and sliding, the gear pair can be modeled as a pair of simple elements in pure rolling contact from the point of view of investigating gross kinematic properties.

Rolling contact can be approached in two different ways, depending on the level of detail desired. If the velocities and accelerations of the rolling elements themselves are immaterial, it is possible to solve for the velocities and accelerations of the other links in a